

Murphy NH₃(g) Data

$$\square A = 27.31$$

$$\square B = 0.02383$$

$$\square C = 1.707 \cdot 10^{-5}$$

$$\square D = -1.185 \cdot 10^{-8}$$

$$\square C_p = A + B T + C T^2 + D T^3$$

$$\triangle C_p = -1.185 \frac{T^3}{10^8} + 1.707 \frac{T^2}{10^5} + 0.02383 T + 27.31 \quad \text{Substitute}$$

$$\square \Delta H = \frac{1}{1000} \left(\int_{298}^{473} C_p dT \right)$$

$$\triangle \Delta H = \frac{1}{1000} \left(-0.29625 \frac{473^4}{10^8} + 0.569 \frac{473^3}{10^5} + 0.29625 \frac{298^4}{10^8} - 0.569 \frac{298^3}{10^5} + 0.011915 \cdot 473^2 - 0.011915 \cdot 298^2 + 4779.3 \right) \quad \text{Substitute}$$

$$\triangle \Delta H = 6.7135 \quad \text{Calculate}$$

Murphy O₂(g) Data

$$\square A = 28.11$$

$$\square B = -3.68 \cdot 10^{-6}$$

$$\square C = 1.746 \cdot 10^{-5}$$

$$\square D = -1.065 \cdot 10^{-8}$$

$$\square C_p = A + B T + C T^2 + D T^3$$

$$\triangle C_p = -3.68 \frac{T}{10^6} - 1.065 \frac{T^3}{10^8} + 1.746 \frac{T^2}{10^5} + 28.11 \quad \text{Substitute}$$

$$\square \Delta H = \frac{1}{1000} \left(\int_{298}^{473} C_p dT \right)$$

$$\triangle \Delta H = \frac{1}{1000} \left(-0.26625 \frac{473^4}{10^8} + 0.582 \frac{473^3}{10^5} - 1.84 \frac{473^2}{10^6} + 0.26625 \frac{298^4}{10^8} - 0.582 \frac{298^3}{10^5} + 1.84 \frac{298^2}{10^6} + 4919.2 \right)$$

$$\triangle \Delta H = 5.2686 \quad \text{Calculate}$$

Murphy O2(g) Data

$$\square A = 28.11$$

$$\square B = -3.68 \cdot 10^{-6}$$

$$\square C = 1.746 \cdot 10^{-5}$$

$$\square D = -1.065 \cdot 10^{-8}$$

$$\square C_p = A + B T + C T^2 + D T^3$$

$$\triangle C_p = -3.68 \frac{T}{10^6} - 1.065 \frac{T^3}{10^8} + 1.746 \frac{T^2}{10^5} + 28.11 \quad \text{Substitute}$$

$$\square \Delta H = \frac{1}{1000} \left(\int_{298}^T C_p dT \right)$$

$$\triangle \Delta H = \frac{1}{1000} \left(-0.26625 \frac{T^4}{10^8} + 0.582 \frac{T^3}{10^5} - 1.84 \frac{T^2}{10^6} + 28.11 T + 0.26625 \frac{298^4}{10^8} - 0.582 \frac{298^3}{10^5} + 1.84 \frac{298^2}{10^6} - 8376.8 \right)$$

$$\triangle \Delta H = 0.001 \left(-2.6625 \times 10^{-9} T^4 + 5.82 \times 10^{-6} T^3 - 1.84 \times 10^{-6} T^2 + 28.11 T - 8509.6 \right) \quad \text{Calculate}$$

$$\triangle \Delta H = -2.6625 \times 10^{-12} T^4 + 5.82 \times 10^{-9} T^3 - 1.84 \times 10^{-9} T^2 + 0.02811 T - 8.5096 \quad \text{Expand}$$

$$\square A = 29.35$$

$$\square B = -9.378 \cdot 10^{-4}$$

$$\square C = 9.747 \cdot 10^{-6}$$

$$\square D = -4.187 \cdot 10^{-9}$$

$$\square C_p = A + B T + C T^2 + D T^3$$

$$\triangle C_p = -4.187 \frac{T^3}{10^9} + 9.747 \frac{T^2}{10^6} - 0.0009378 T + 29.35 \quad \text{Substitute}$$

$$\square \Delta H = \frac{1}{1000} \left(\int_{298}^T C_p dT \right)$$

$$\triangle \Delta H = \frac{1}{1000} \left(-1.0468 \frac{T^4}{10^9} + 3.249 \frac{T^3}{10^6} - 0.0004689 T^2 + 29.35 T + 1.0468 \frac{298^4}{10^9} - 3.249 \frac{298^3}{10^6} + 0.0004689 \cdot 298^2 - 8746.3 \right) \quad \text{Substit}$$

$$\triangle \Delta H = 0.001 \left(-1.0468 \times 10^{-9} T^4 + 3.249 \times 10^{-6} T^3 - 0.0004689 T^2 + 29.35 T - 8782.4 \right) \quad \text{Calculate}$$

$$\triangle \Delta H = -1.0468 \times 10^{-12} T^4 + 3.249 \times 10^{-9} T^3 - 4.689 \times 10^{-7} T^2 + 0.02935 T - 8.7824 \quad \text{Expand}$$

Murphy NO(g) Data

Murphy H₂O(g) Data

☐ $A = 32.24$

☐ $B = 0.001924$

☐ $C = 1.005 \cdot 10^{-5}$

☐ $D = -3.59 \cdot 10^{-9}$

☐ $C_p = A + B T + C T^2 + D T^3$

$\Delta C_p = -3.59 \frac{T^3}{10^9} + 1.005 \frac{T^2}{10^5} + 0.001924 T + 32.24$ *Substitute*

☐ $\Delta H = \frac{1}{1000} \left(\int_{298}^T C_p dT \right)$

$\Delta H = \frac{1}{1000} \left(-0.8975 \frac{T^4}{10^9} + 0.335 \frac{T^3}{10^5} + 0.000962 T^2 + 32.24 T + 0.8975 \frac{298^4}{10^9} - 0.335 \frac{298^3}{10^5} - 0.000962 \cdot 298^2 - 9607.5 \right)$

$\Delta H = 0.001 \left(-8.975 \times 10^{-10} T^4 + 3.35 \times 10^{-6} T^3 + 0.000962 T^2 + 32.24 T - 9774.5 \right)$ *Calculate*

$\Delta H = -8.975 \times 10^{-13} T^4 + 3.35 \times 10^{-9} T^3 + 9.62 \times 10^{-7} T^2 + 0.03224 T - 9.7745$ *Expand*

Energy Balance

☐ $\Delta H_a = 6.7135$

☐ $\Delta H_b = 5.2686$

☐ $\Delta H_c = -2.6625 \times 10^{-12} T^4 + 5.82 \times 10^{-9} T^3 - 1.84 \times 10^{-9} T^2 + 0.02811 T - 8.5096$

☐ $\Delta H_d = -1.0468 \times 10^{-12} T^4 + 3.249 \times 10^{-9} T^3 - 4.689 \times 10^{-7} T^2 + 0.02935 T - 8.7824$

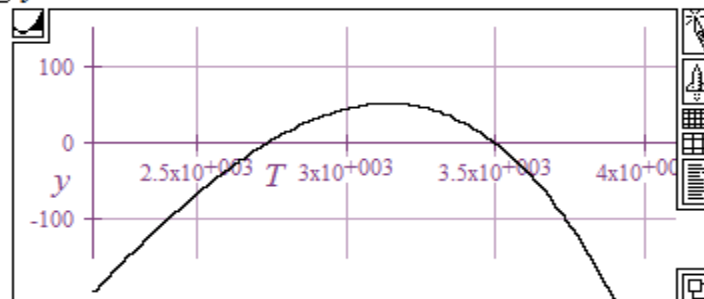
☐ $\Delta H_e = -8.975 \times 10^{-13} T^4 + 3.35 \times 10^{-9} T^3 + 9.62 \times 10^{-7} T^2 + 0.03224 T - 9.7745$

☐ $0 = 1(-905.9) + 1 \Delta H_c + 4 \Delta H_d + 6 \Delta H_e - 4 \Delta H_a - 6 \Delta H_b$

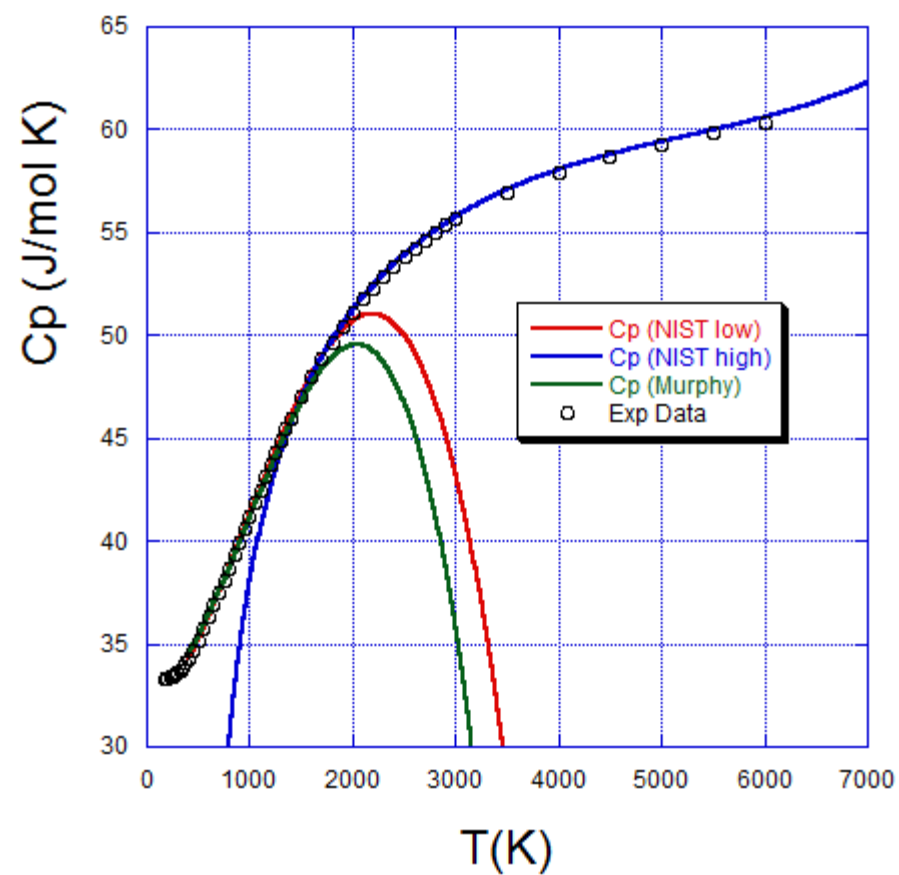
$\Delta 0 = 6 \left(-8.975 \times 10^{-13} T^4 + 3.35 \times 10^{-9} T^3 + 9.62 \times 10^{-7} T^2 + 0.03224 T - 9.7745 \right) + 4 \left(-1.0468 \times 10^{-12} T^4 + 3.249 \times 10^{-9} T^3 - 4.689 \times 10^{-7} T^2 + 0.02935 T - 8.7824 \right) - 4(6.7135) - 6(5.2686)$

$\Delta 0 = -1.2234 \times 10^{-11} T^4 + 3.8916 \times 10^{-8} T^3 + 3.8946 \times 10^{-6} T^2 + 0.33895 T - 1066.7$

☒ $y = -1.2234 \times 10^{-11} T^4 + 3.8916 \times 10^{-8} T^3 + 3.8946 \times 10^{-6} T^2 + 0.33895 T - 1066.7$



Heat Capacity of Water Vapor NIST vs. Murphy



NIST Oxygen (O2) Data

Gas Phase Heat Capacity (Shomate Equation)

$$C_p^\circ = A + B \cdot t + C \cdot t^2 + D \cdot t^3 + E/t^2$$

$$H^\circ - H_{298.15}^\circ = A \cdot t + B \cdot t^2/2 + C \cdot t^3/3 + D \cdot t^4/4 - E/t + F - H$$

$$S^\circ = A \cdot \ln(t) + B \cdot t + C \cdot t^2/2 + D \cdot t^3/3 - E/(2 \cdot t^2) + G$$

C_p = heat capacity (J/mol*K)

H° = standard enthalpy (kJ/mol)

S° = standard entropy (J/mol*K)

t = temperature (K) / 1000.

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[View table.](#)

Temperature (K)	100. - 700.	700. - 2000.	2000. - 6000.
A	31.32234	30.03235	20.91111
B	-20.23531	8.772972	10.72071
C	57.86644	-3.988133	-2.020498
D	-36.50624	0.788313	0.146449
E	-0.007374	-0.741599	9.245722
F	-8.903471	-11.32468	5.337651
G	246.7945	236.1663	237.6185
H	0.0	0.0	0.0
Reference	Chase, 1998	Chase, 1998	Chase, 1998
Comment	Data last reviewed in March, 1977; New parameter fit January 2009	Data last reviewed in March, 1977; New parameter fit January 2009	Data last reviewed in March, 1977; New parameter fit January 2009

NIST Nitric Oxide (NO) Data

Gas Phase Heat Capacity (Shomate Equation)

$$C_p^\circ = A + B \cdot t + C \cdot t^2 + D \cdot t^3 + E/t^2$$

$$H^\circ - H^\circ_{298.15} = A \cdot t + B \cdot t^2/2 + C \cdot t^3/3 + D \cdot t^4/4 - E/t + F - H$$

$$S^\circ = A \cdot \ln(t) + B \cdot t + C \cdot t^2/2 + D \cdot t^3/3 - E/(2 \cdot t^2) + G$$

C_p = heat capacity (J/mol*K)

H° = standard enthalpy (kJ/mol)

S° = standard entropy (J/mol*K)

t = temperature (K) / 1000.

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[View table.](#)

Temperature (K)	298. - 1200.	1200. - 6000.
A	23.83491	35.99169
B	12.58878	0.957170
C	-1.139011	-0.148032
D	-1.497459	0.009974
E	0.214194	-3.004088
F	83.35783	73.10787
G	237.1219	246.1619
H	90.29114	90.29114
Reference	Chase, 1998	Chase, 1998
Comment	Data last reviewed in June, 1963	Data last reviewed in June, 1963

NIST Water Vapor Data

Gas Phase Heat Capacity (Shomate Equation)

$$C_p^\circ = A + B \cdot t + C \cdot t^2 + D \cdot t^3 + E/t^2$$

$$H^\circ - H^\circ_{298.15} = A \cdot t + B \cdot t^2/2 + C \cdot t^3/3 + D \cdot t^4/4 - E/t + F - H$$

$$S^\circ = A \cdot \ln(t) + B \cdot t + C \cdot t^2/2 + D \cdot t^3/3 - E/(2 \cdot t^2) + G$$

C_p = heat capacity (J/mol*K)

H° = standard enthalpy (kJ/mol)

S° = standard entropy (J/mol*K)

t = temperature (K) / 1000.

[View plot](#) Requires a JavaScript / HTML 5 canvas capable browser.

[View table.](#)

Temperature (K)	500. - 1700.	1700. - 6000.
A	30.09200	41.96426
B	6.832514	8.622053
C	6.793435	-1.499780
D	-2.534480	0.098119
E	0.082139	-11.15764
F	-250.8810	-272.1797
G	223.3967	219.7809
H	-241.8264	-241.8264
Reference	Chase, 1998	Chase, 1998
Comment	Data last reviewed in March, 1979	Data last reviewed in March, 1979

$$\square H_a = 6.7048$$

$$\square H_b = 5.253$$

$$\square H_c = -9.2457\frac{1}{t} + 0.036612t^4 - 0.6735t^3 + 5.3604t^2 + 20.911t + 5.3377$$

$$\square H_d = 3.0041\frac{1}{t} + 0.0024935t^4 - 0.049343t^3 + 0.47858t^2 + 35.992t - 17.183$$

$$\square H_e = 11.158\frac{1}{t} + 0.02453t^4 - 0.49993t^3 + 4.311t^2 + 41.964t - 30.354$$

$$\square 0 = -905.9 + (1H_c + 4H_d + 6H_e) - (4H_a + 6H_b)$$

$$\triangle 0 = 4\left(3.0041\frac{1}{t} + 0.0024935t^4 - 0.049343t^3 + 0.47858t^2 + 35.992t - 17.183\right)$$

$$+ 6\left(11.158\frac{1}{t} + 0.02453t^4 - 0.49993t^3 + 4.311t^2 + 41.964t - 30.354\right) - 9.2457\frac{1}{t} + 0.036612t^4 - 0.6735t^3 + 5.3604t^2 + 20.911t - 958.9$$

$$\triangle 0 = 69.716\frac{1}{t} + 0.19376t^4 - 3.8704t^3 + 33.141t^2 + 416.66t - 1209.8 \quad \text{Expand}$$

$$\blacksquare y = 69.716\frac{1}{t} + 0.19376t^4 - 3.8704t^3 + 33.141t^2 + 416.66t - 1209.8$$

